

Heart Murmur Detection – Complete Notebook Explanation

This notebook builds a **heart sound classification system** using **audio preprocessing + LSTM deep learning**.

The goal is to classify heart sounds into **Artifact, Murmur, and Normal (Others)** and test the model on **unlabelled heart sound data**.

1 Data Loading Phase

("Getting the raw data into the system")

What kind of data do we have?

- **Heart sound recordings (.wav files)**
- Different categories:
 - Artifact
 - Murmur
 - Normal
 - Extrahls
 - Extrastole
- **Plus unlabelled test audio**

The heart sound categories in your dataset represent different auditory patterns identified during a cardiac examination, ranging from healthy sounds to various abnormalities and technical noise.

Here is an explanation of each sound category:

Normal: These are typical, healthy heart sounds, often described as a clear and rhythmic "lub-dub". The "lub" (S1) is the sound of the mitral and tricuspid valves closing at the beginning of the heart's contraction phase (systole), and the "dub" (S2) is the sound of the aortic and pulmonary valves closing at the beginning of the relaxation phase (diastole).

Murmur: This is an extra, distinct sound, often described as a "whooshing," "swishing," "blowing," or "rumbling" noise, caused by turbulent or rapid blood flow through the heart. Murmurs can be harmless (innocent) or a sign of an underlying heart problem, such as a problem with a heart valve (e.g., stenosis or regurgitation). The sound occurs between, after, or sometimes encompasses the normal S1 and S2 sounds.

Extrahls (Extra Heart Sounds): This refers to the presence of additional heart sounds beyond the normal S1 and S2, most commonly the third heart sound (S3) or fourth heart sound (S4). An S3 sound may be normal in children and young adults but can indicate issues in older adults (e.g., heart failure). It's a low-pitched sound heard after S2, sometimes creating a "galloping" rhythm.

An S4 sound is almost always a sign of a pathological condition, caused by a stiff ventricle that forces the atrium to contract harder to push blood in, heard just before S1 (a "ta-lub-dub" sound).

Extrastole: This refers to an out-of-rhythm heartbeat, typically a premature contraction of the heart (an extra or skipped beat). The heart sound pattern is disrupted by this premature event, leading to an irregular rhythm.

Artifact: These are sounds present in the recording that are not physiological heart sounds. Artifacts are essentially noise, which can include a wide range of different sounds such as background speech, music, stethoscope feedback squeals, or other non-cardiac noise, which interfere with the analysis of the actual heartbeat.

Unlabelled test audio: These are audio files that have not been assigned a category and are likely intended to be used to test the performance of a machine learning model or diagnostic system after it has been trained on the labelled data.

Why load data class-wise first?

We load files **separately per class** because:

- It helps us **assign correct labels**
- Makes it easy to **inspect class imbalance**
- Keeps the dataset **organized and readable**

Example:

```
artifact_sounds → label 0  
murmur_sounds  → label 1  
others          → label 2
```

👉 At this stage:

- No DL
- No feature extraction
- Just **raw audio signals**

2 Preprocessing Phase (MOST IMPORTANT PART)

("Making audio understandable for a machine")

Audio cannot be directly fed into ML models.

So we **convert raw sound into meaningful numerical features**.

◆ Step 1: Fix Sampling Rate

`SAMPLE_RATE = 22050`

What is the sampling rate?

- Number of audio samples captured **per second**
- 22,050 Hz = 22,050 values per second

Why do this?

- Audio files come with **different sampling rates**
- ML models need **uniform data**
- 22 kHz is:
 - Standard
 - Light on computation
 - Enough to capture heart sounds

◆ Step 2: Fix Audio Duration

`MAX_SOUND_CLIP_DURATION = 10`

What does this mean?

- Every audio clip is forced to be **10 seconds**
- If shorter → padded
- If longer → trimmed

Why is this critical?

- LSTM requires **same-length sequences**
 - Variable length = model crash
 - Ensures **consistent input shape**
-

◆ Step 3: Why we don't train on raw audio

Raw waveform looks like:

`[0.02, -0.01, 0.03, ...]`

Problems:

- Too large
- Too noisy
- No meaningful structure for learning

👉 So we convert audio → **features**

3 Feature Extraction

(“Turning sound into learnable patterns”)

This is the heart of preprocessing.

◆ What is a Spectrum?

- Shows which frequencies exist in sound
- Tells:
 - Low-frequency heart beats
 - High-frequency murmurs

But spectrum alone loses time information.

◆ What is a Spectrogram?

- A 2D image
- X-axis → time
- Y-axis → frequency
- Color → strength

This shows:

- When murmurs occur
- How sound changes over time

Still heavy for sequence models.

♦ What is MFCC? (Very Important)

MFCC = Mel Frequency Cepstral Coefficients

In simple words:

MFCCs are compact numerical features that describe how sound actually sounds to human ears.

Why MFCCs?

- Removes unnecessary noise
 - Focuses on important frequency patterns
 - Widely used in:
 - Speech recognition
 - Heart sound analysis
 - Much smaller than spectrograms
-

What MFCC captures:

- Heart beat rhythm
- Murmur texture
- Frequency distribution over time

This makes MFCC perfect for LSTM.

Handling Unlabelled Data

("Testing on unseen heart sounds")

Unlabelled data:

- Has **no class information**
- Assigned label **-1**

Why still preprocess it?

- Model expects **same input format**
- Ensures:
 - Same sampling rate
 - Same duration
 - Same MFCC extraction

 Unlabelled data is:

- NOT used in training
- ONLY used for prediction

This avoids **data leakage**.

5 Dataset Combination

("Preparing data for model input")

What we do:

- Combine all labeled audio into:

`x_data → audio signals`

`y_data → class labels`

- Combine unlabelled audio into:

`test_x → unseen data`

Why?

DL models expect:

`X → all samples`

`y → all labels`

6 Dataset Imbalance Problem

("Real-world medical data issue")

Observed:

- Normal samples >> Murmur >> Artifact

This is dangerous because:

- Model may predict **Normal always**
- Minority classes get ignored

7 Class Weighting (Smart Balancing)

Instead of duplicating data, we use **class weights**.

What are class weights?

- Higher penalty for minority class errors
- Lower penalty for majority class errors

Why is this the best?

- Keeps data original
- Prevents overfitting
- Preferred in **medical ML**

👉 Balancing happens in **loss function**, not data size.

8 Model Training Phase

("Teaching the model patterns")

Why LSTM?

Heart sounds are:

- Sequential
- Time-dependent
- Pattern-based

LSTM:

- Remembers past sounds
- Detects rhythm changes
- Ideal for murmurs

Training process:

- Input → MFCC sequences
- Output → class labels
- Loss → weighted (class_weight)
- Optimizer → improves model step by step

9 Model Evaluation

("Checking how good the model is")

Evaluation includes:

- Accuracy
- Confusion matrix
- Predictions on unlabelled data

Key focus:

- Murmur detection recall
- Minority class performance

10 Final End-to-End Flow

```
Heart sound (.wav)
↓
Resample to 22 kHz
↓
Fix duration (10 sec)
↓
Extract MFCC features
↓
Handle imbalance using class weights
↓
Train LSTM model
↓
Evaluate & predict on unseen data
```

SO summary is ,

“We convert heart sound recordings into MFCC features, train an LSTM model using class-weighted learning to handle imbalance, and evaluate the model on both labeled and unseen heart sounds.”